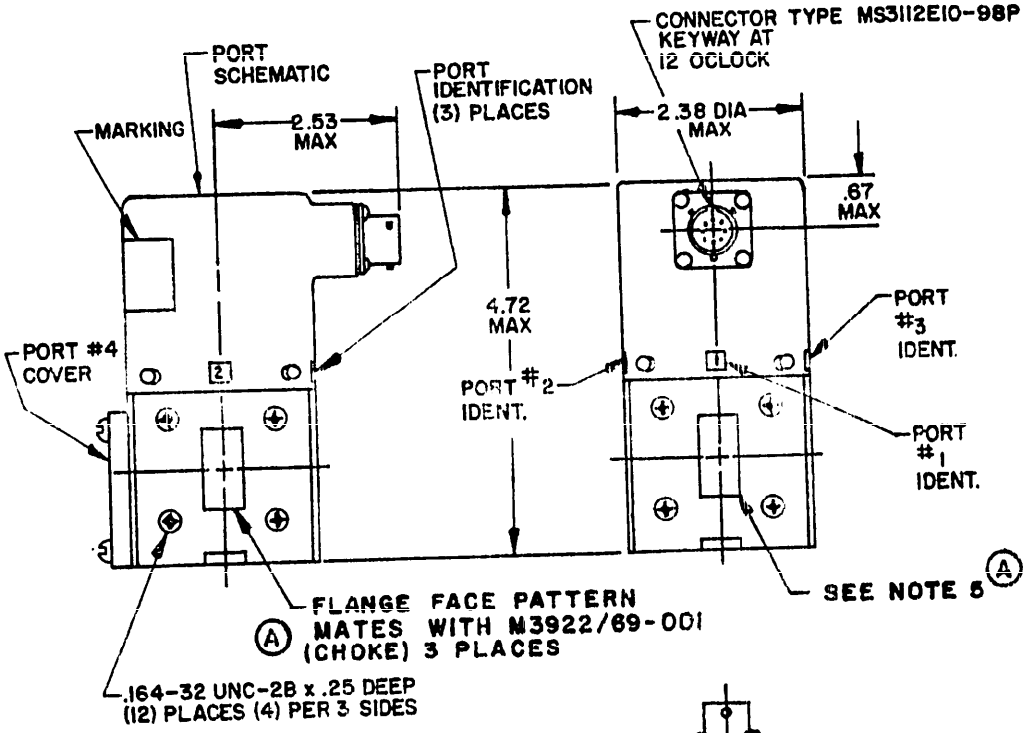
MILITARY SPECIFICATION SHEET

SWITCHES, WAVEGUIDE, SP2T, FAIL-SAFE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-55041.



INCHES	MM
.164	4.17
.195	4.95
.235	5.9
.25	6.35
.31	7.9
.67	17.01
1.000	25.4
1.156	29.4
2.000	50.8
2.312	58.7
2.38	60.4
2.53	64.3
4.72	119.8

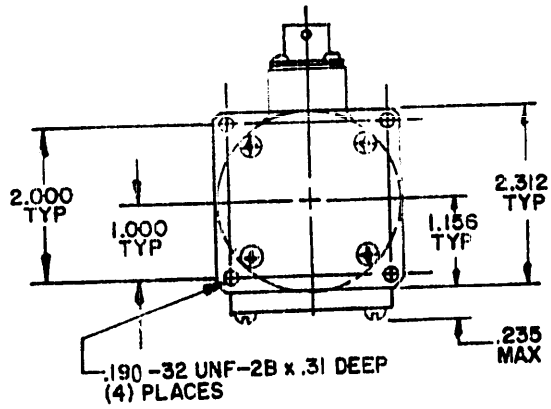


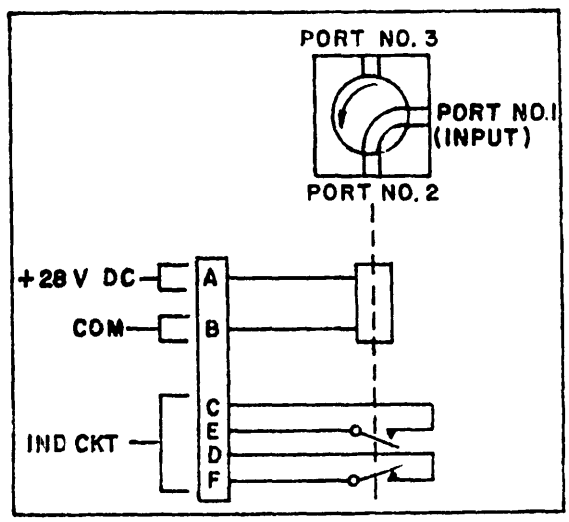
FIGURE 1. Switch configuration and schematic, part no. M55041/3-001.

(A) denotes changes

FSC 5985

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SCHEMATIC

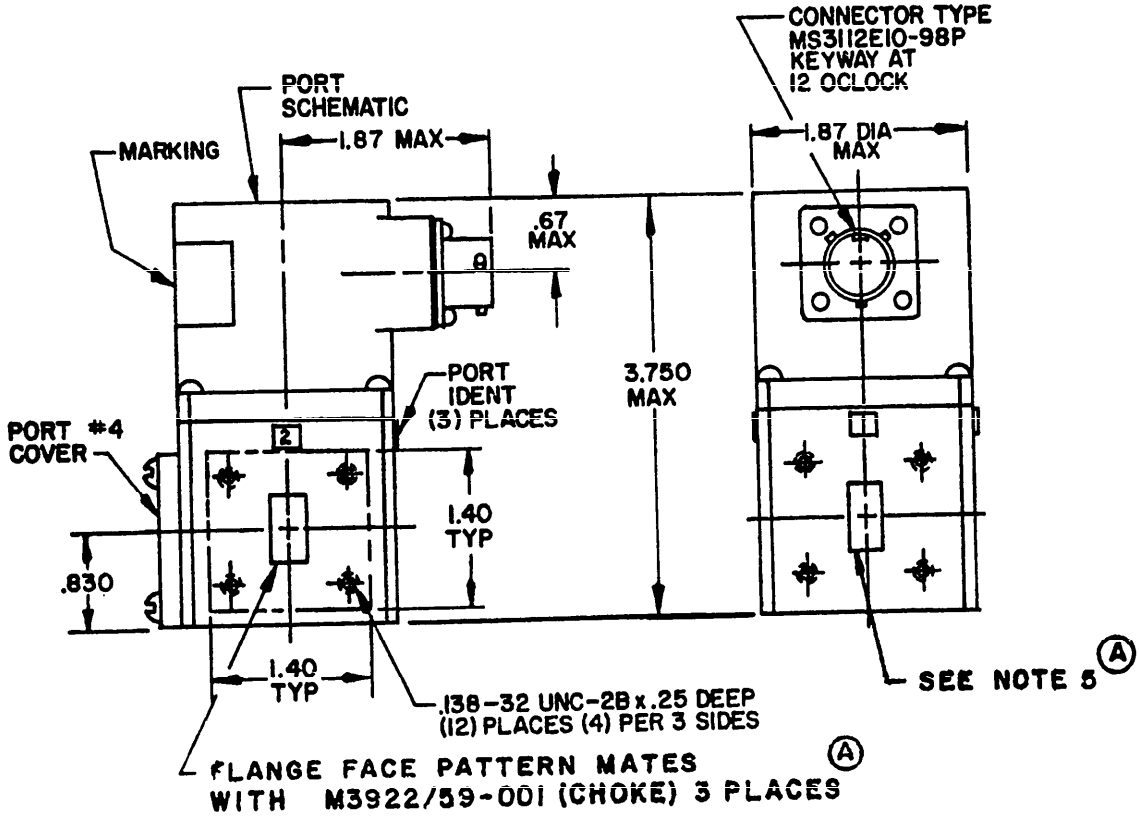


SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-155.
Ⓐ 6. DC polarity must be observed.

FIGURE 1. Switch configuration and schematic, part no. M55041/3-001 - Continued.

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INCHES	MM
.138	3.5
.164	4.17
.235	5.9
.25	6.35
.67	17.01
.718	18.2
.830	21.08
.938	23.8
1.40	35.5
1.437	36.5
1.65	41.9
1.87	47.5
1.875	47.6
3.750	95.2

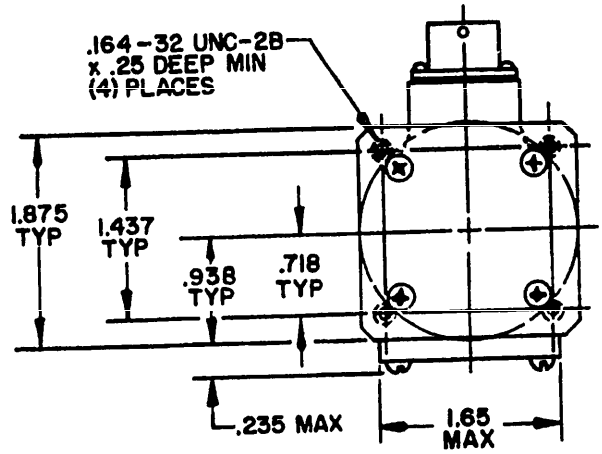
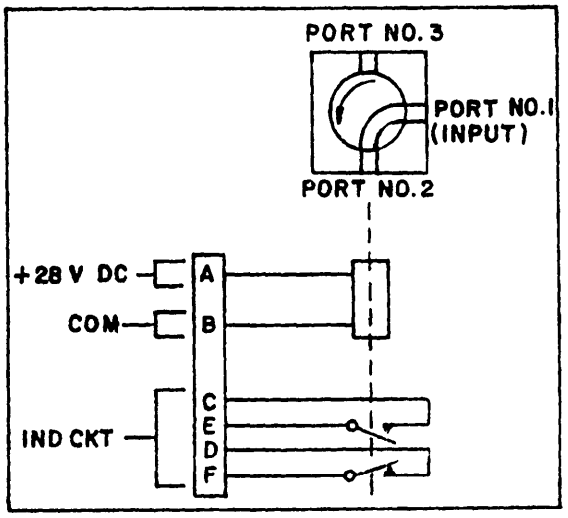


FIGURE 2. Switch configuration and schematic, part no. M55041/3-002.

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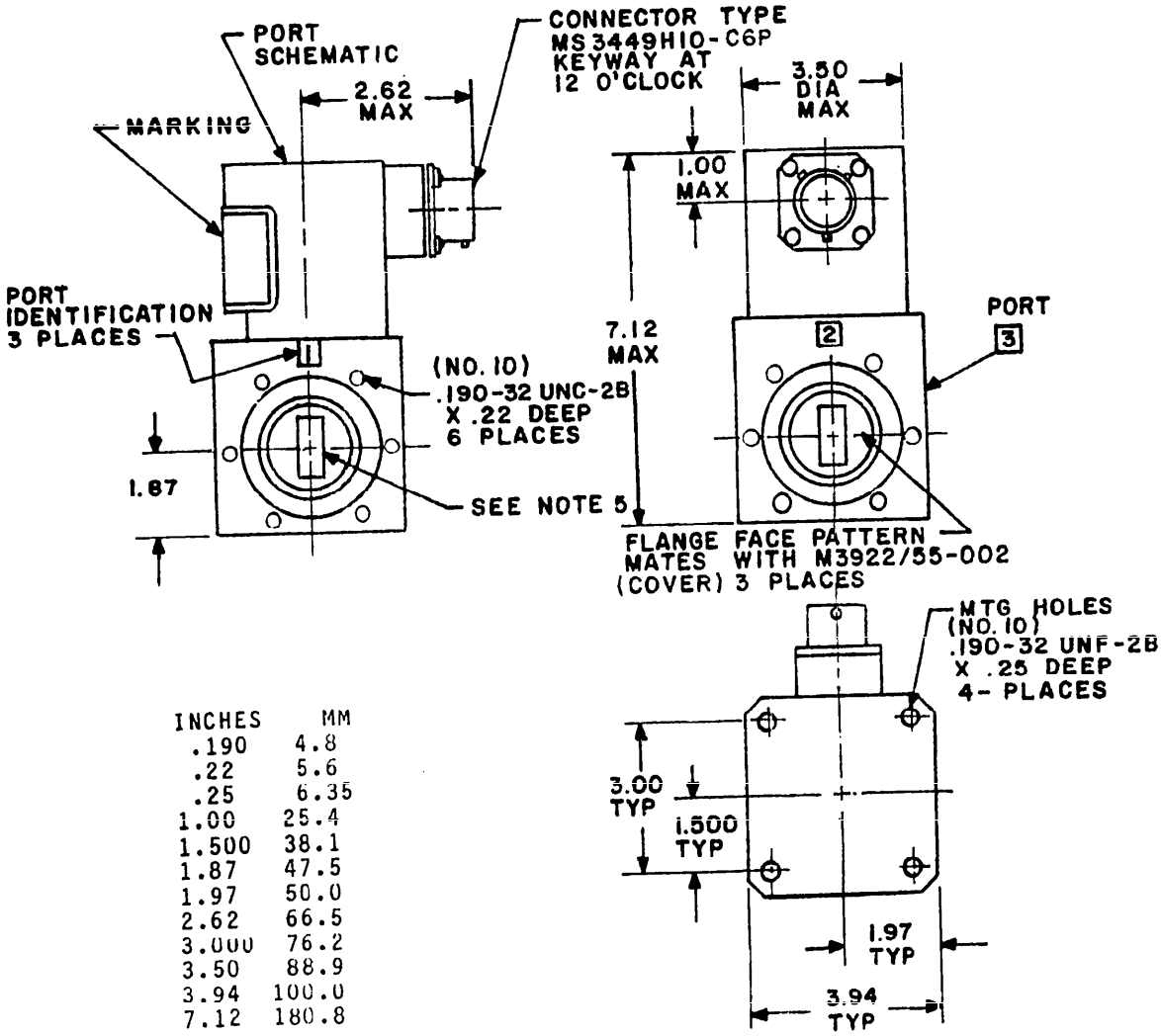


SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-093.
Ⓐ 6. DC polarity must be observed.

FIGURE 2. Switch configuration and schematic, part no. M55041/3-002 - Continued.

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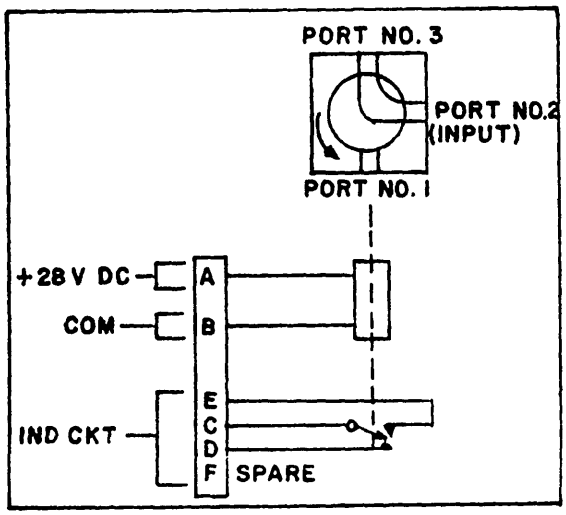


INCHES	MM
.190	4.8
.22	5.6
.25	6.35
1.00	25.4
1.500	38.1
1.87	47.5
1.97	50.0
2.62	66.5
3.000	76.2
3.50	88.9
3.94	100.0
7.12	180.8

Ⓐ FIGURE 3. Switch configuration and schematic, part no. M55041/3-003.

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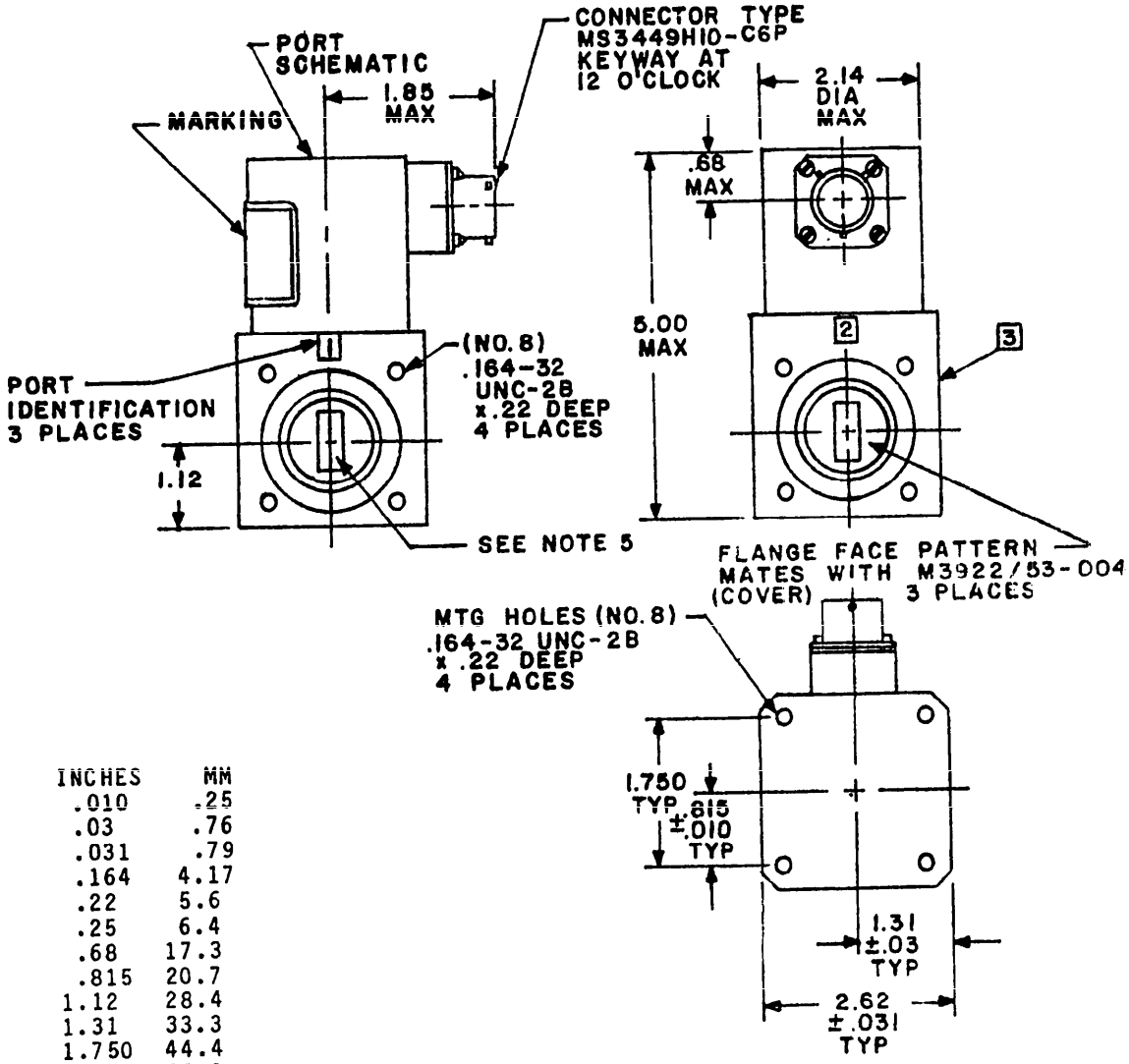


SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-065.
Ⓐ 6. DC polarity must be observed.

FIGURE 3. Switch configuration and schematic, part no. M55041/3-003 - Continued.

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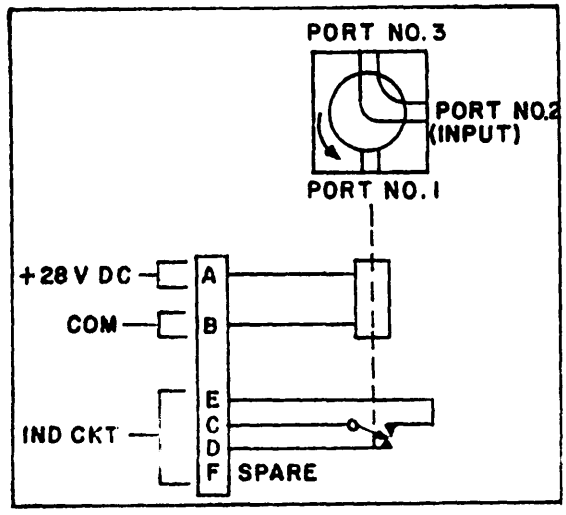


INCHES	MM
.010	.25
.03	.76
.031	.79
.164	4.17
.22	5.6
.25	6.4
.68	17.3
.815	20.7
1.12	28.4
1.31	33.3
1.750	44.4
1.85	46.9
2.14	54.4
2.62	66.5
5.00	127.0

Ⓐ FIGURE 4. Switch configuration and schematic, part no. M55041/3-004.

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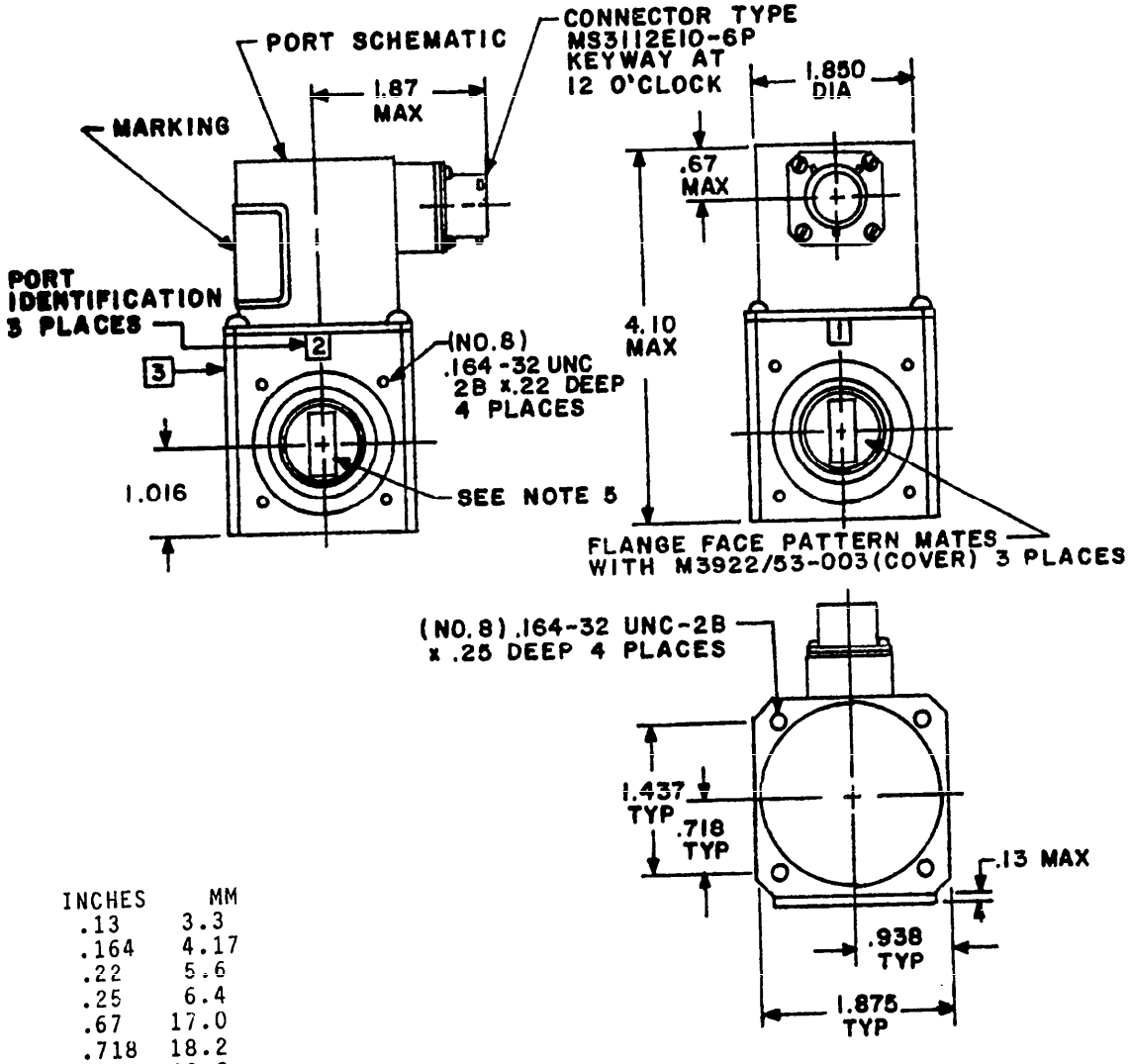


SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-072.
Ⓐ 6. DC polarity must be observed.

FIGURE 4. Switch configuration and schematic, part no. M55041/3-004 - Continued.

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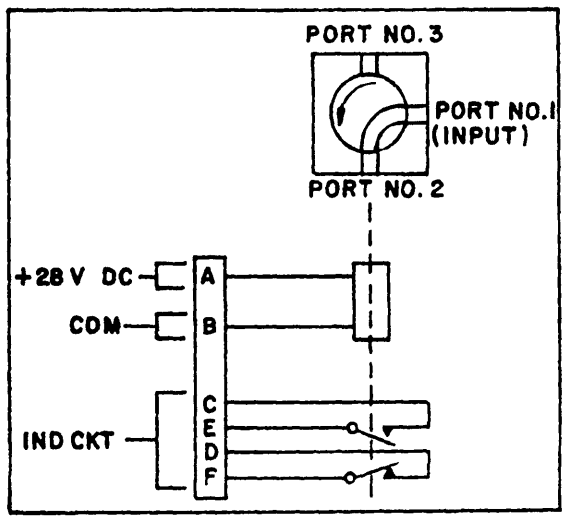


INCHES	MM
.13	3.3
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.718	18.2
.938	23.8
1.014	25.8
1.437	36.5
1.850	46.9
1.87	47.5
1.875	47.6
4.10	104.14

Ⓐ FIGURE 5. Switch configuration and schematic, part no. M55041/3-005.

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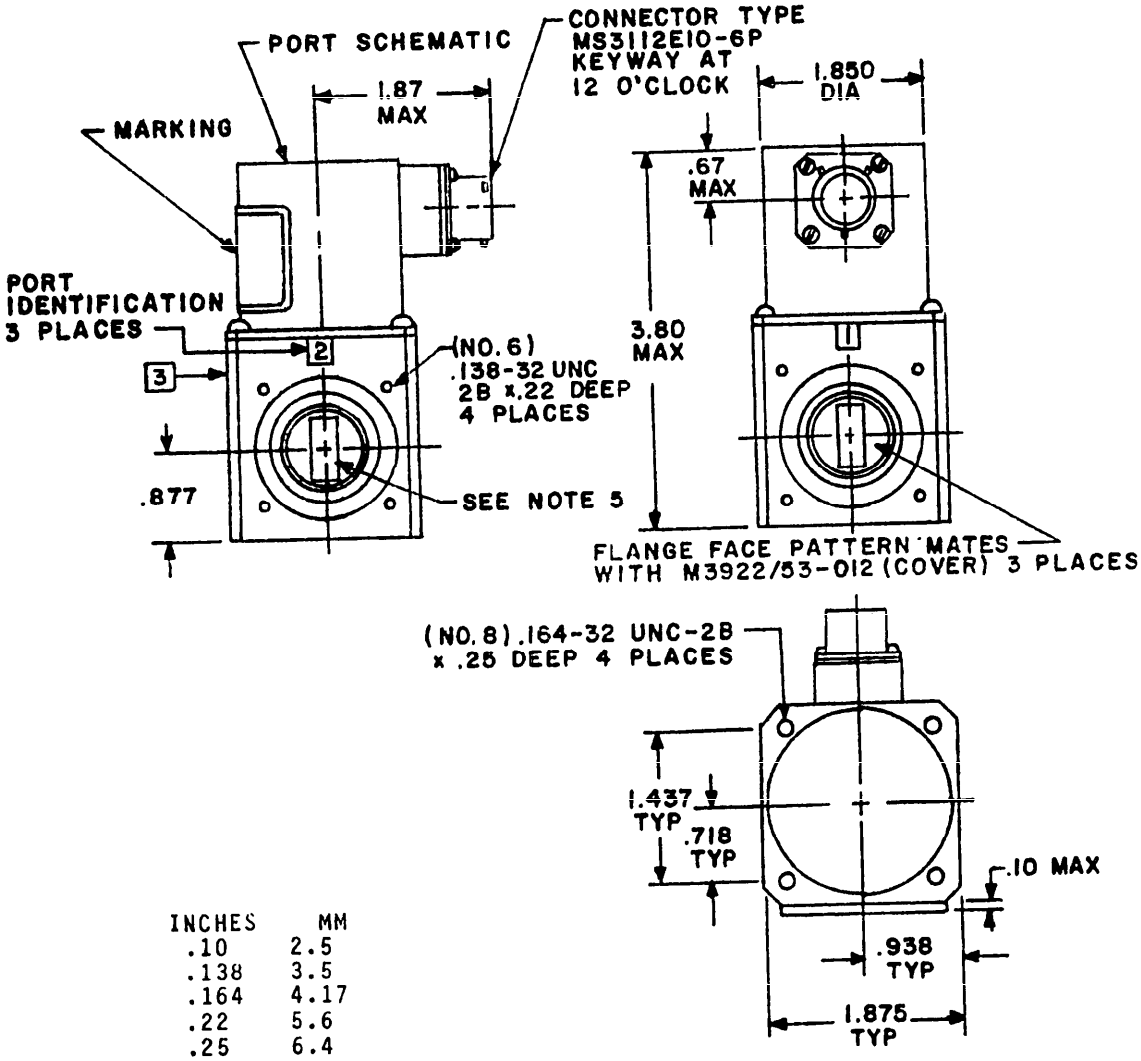


SHOWN IN DEENERGIZED POSITION (POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-078.
Ⓐ 6. DC polarity must be observed.

FIGURE 5. Switch configuration and schematic, part no. M55041/3-005 - Continued.

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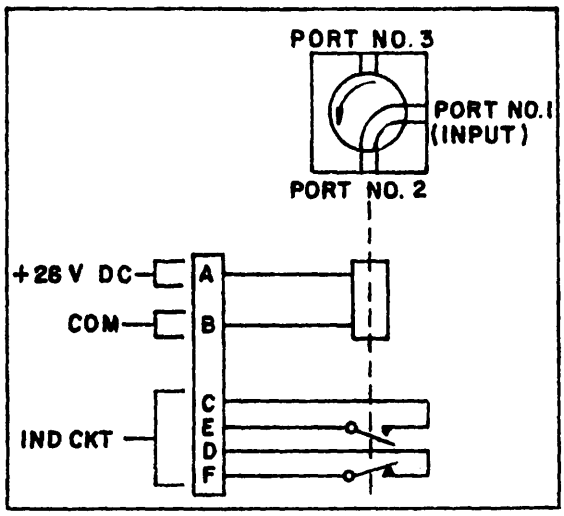


INCHES	MM
.10	2.5
.138	3.5
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.718	18.2
.877	22.3
.938	23.8
1.437	36.5
1.850	46.9
3.80	96.5

Ⓐ FIGURE 6. Switch configuration and schematic, part no. M55041/3-006.

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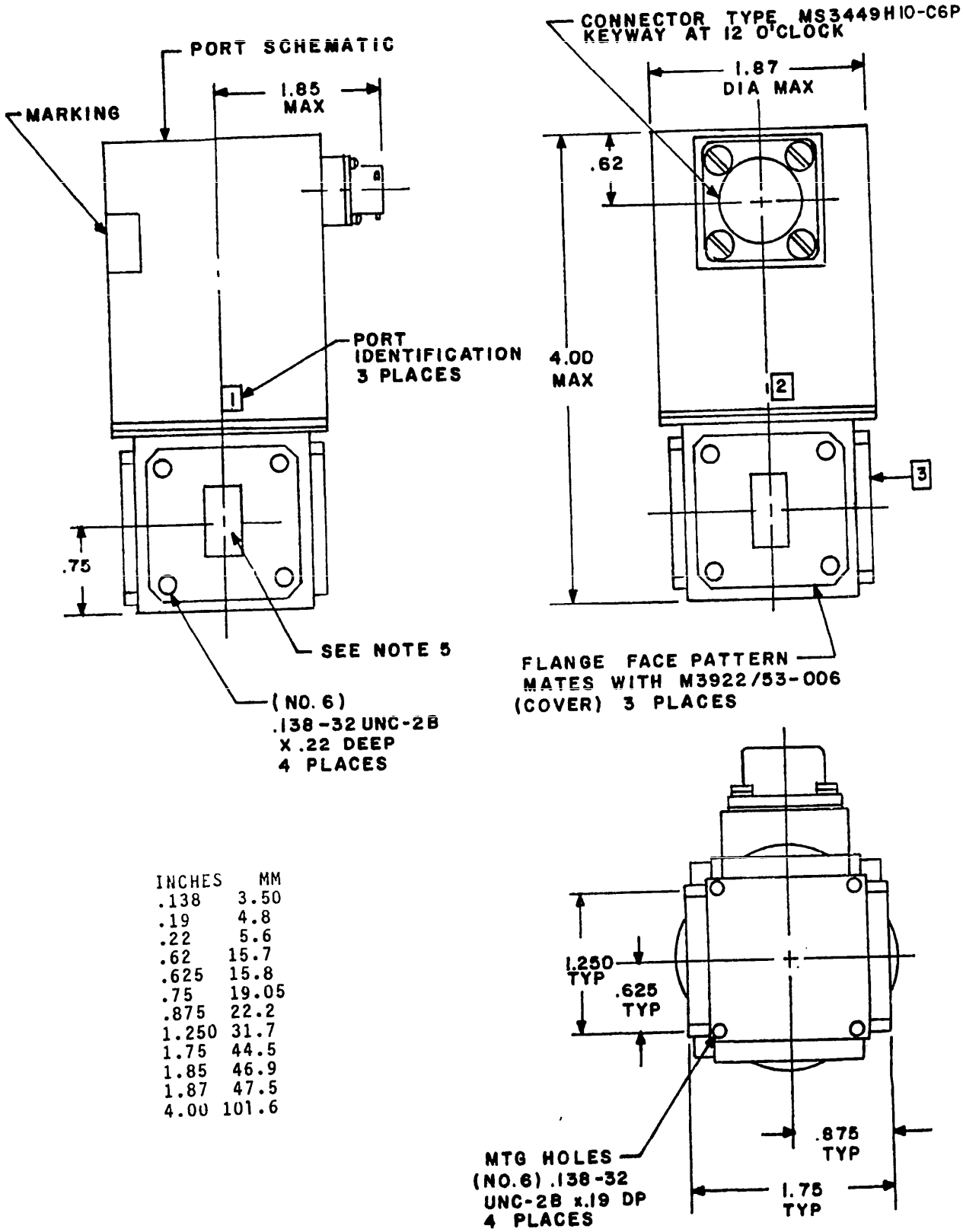


SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
Ⓐ 6. DC polarity must be observed.

FIGURE 6. Switch configuration and schematic, part no. M55041/3-006 - Continued.

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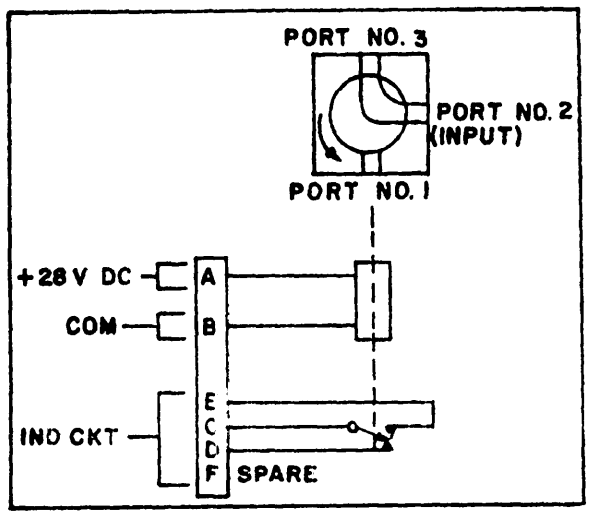


INCHES	MM
.138	3.50
.19	4.8
.22	5.6
.62	15.7
.625	15.8
.75	19.05
.875	22.2
1.250	31.7
1.75	44.5
1.85	46.9
1.87	47.5
4.00	101.6

Ⓐ FIGURE 7. Switch configuration and schematic, part no. M55041/3-007.

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SCHEMATIC



SHOWN IN DEENERGIZED POSITION (POSITION I)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
Ⓐ 6. DC polarity must be observed.

FIGURE 7. Switch configuration and schematic, part no. M55041/3-007 - Continued.

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Ⓐ TABLE I. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Position indicator circuit	Inter-lock circuit	Nominal operating voltage	Pull-in voltage (less than)	Dropout voltage (less than)	Operating current (max) 3/	Hold-ing current (max)
			GHz	3/ amperes		V dc	V dc	V dc	amperes	amperes
001	1	1/ 2/	7.0-11.25	2	N/A	28	20	10	1.5	0.5
002	2	1/ 2/	11.25-18.00	2	N/A	28	20	10	1.0	0.5
003	3	1/ 2/	5.8-8.2	3	N/A	28	20	10	1.25	0.5
004	4	1/ 2/	7.05-10	3	N/A	28	20	10	1.25	0.5
005	5	1/ 2/	8.2-12.4	3	N/A	28	20	10	1.0	1.0
006	6	1/ 2/	12.4-18.0	3	N/A	28	20	10	0.5	0.5
007	7	1/ 2/	12.4-18.0	3	N/A	28	20	10	1.2	0.5

TABLE I. Electrical performance characteristics - Continued.

Dash number	RF power handling capability	Isolation (min)	VSWR (max)	Insertion loss (max)	Switching time	Waveguide pressuri-zation	Life cycles (X1000)	Weight (max)
		dB		dB	ms	lb_f/in^2		oz
001	250 W	60	1.15:1	0.2	100	20	100	46
002	100 W	60	1.15:1	0.2	100	20	100	21
003	560 kW (peak) 4/	70	1.05:1	0.05	125	45	500	49
004	350 kW (peak) 5/	70	1.10:1	0.05	75	45	500	44
005	1200 W	60	1.10:1	0.1	100	20	200	23
006	750 W	60	1.10:1	0.1	100	20	200	21
007	120 kW (peak) 7/	70	1.10:1	0.1	75	45	500	23

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.

2/ Rotor and housing of aluminum alloy.

3/ At 28 V dc.

4/ Duty cycle of 26×10^{-3}.

5/ Duty cycle of 27×10^{-3}.

6/ Duty cycle of 24×10^{-3}.

7/ Duty cycle of 1×10^{-3}.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures.

Power connector: See figures.

Operation: Break before make.

Operating temperature range: -55°C to +95°C.

Electrical performance characteristics: See table I.

Part number: M55041/3- (and dash number from table I).

Ⓐ Substitutability data: The switches covered by this specification sheet are substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985-
001	07397	20-811079-2	00-006-2054
001	82152	35D02800	---
001	24402	WS-02Z11-3	00-006-2054
002	07397	20-811079-1	00-006-2055
002	82152	32C01200	---
002	24402	WS-02P11-15	00-006-2055
003	09017	600961-01	---
003	01456	137SR16-45P-2-28	---
004	09017	600960-01	---
004	01456	112SR36-45P-2-28	---
005	93346	3451570-1	---
005	82152	33D01300	---
006	93346	3451570-2	---
006	82152	33D00300	01-039-8434
007	09017	600958-1	---
007	01456	62SR36-45P-2-28	---

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0975-1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-55041/3A	Switches, Waveguide, 2P2T, Fail-safe

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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